# SQL FINANCIAL ANALYSIS PIPELINE

*Project: Tesla (TSLA) Stock Historical Data Analysis • S Data Solutions*

## 1. Table Schema & Data Definition (DDL)

Optimized table structure designed to handle Tesla's historical time-series pricing and large trade volumes.

```sql
CREATE TABLE tesla_stock_data (
    Date DATE,
    Adj_Close DECIMAL(12,6),
    Close DECIMAL(12,6),
    High DECIMAL(12,6),
    Low DECIMAL(12,6),
    Open DECIMAL(12,6),
    Volume BIGINT
);
```

## 2. Data Cleaning & Integrity Control

Validating time-series progression uniqueness and ensuring filtering of record corruption or negative values.

```sql
-- Ensure Date is unique (Primary Key validation)
SELECT Date, COUNT(*) FROM tesla_stock_data GROUP BY Date HAVING COUNT(*) > 1;

-- Flag pricing anomalies, zero entries, or negative market values
SELECT * FROM tesla_stock_data WHERE Open <= 0 OR Close <= 0 OR Volume < 0;
```

## 3. Core Market KPIs (Historical Benchmarks)

Extracting critical historical growth milestones, price extremes, and liquidity baselines.

```sql
SELECT
    MIN(Date) AS market_debut, MAX(Date) AS latest_record,
    MAX(High) AS all_time_high, MIN(Low) AS all_time_low,
    ROUND(AVG(Volume), 0) AS avg_daily_volume
FROM tesla_stock_data;
```

## 4. Business Insights & Annual Stock Performance

Aggregating price benchmarks across tracking years and isolating trading sessions with peak volatility spreads.

```sql
-- Yearly average closing price trend progression
SELECT EXTRACT(YEAR FROM Date) AS trading_year,
       ROUND(AVG(Close), 2) AS average_closing_price
FROM tesla_stock_data GROUP BY EXTRACT(YEAR FROM Date) ORDER BY trading_year DESC;

-- Top 5 trading days highlighting the highest intra-day volatility spread
SELECT Date, Open, Close, High, Low, (High - Low) AS intraday_spread
FROM tesla_stock_data ORDER BY intraday_spread DESC LIMIT 5;
```

## 5. Advanced Analytics (Window Functions)

Computing consecutive day return rates and modeling a 20-day historical Simple Moving Average (SMA).

```sql
-- Daily return rate percentage calculation using LAG()
SELECT Date, Close,
       ROUND(((Close - LAG(Close, 1) OVER(ORDER BY Date)) / LAG(Close, 1) OVER(ORDER BY Date)) * 1
FROM tesla_stock_data;

-- 20-Day Simple Moving Average (SMA) trend calculation
SELECT Date, Close,
       ROUND(AVG(Close) OVER(ORDER BY Date ROWS BETWEEN 19 PRECEDING AND CURRENT ROW), 2) AS movin
FROM tesla_stock_data;
```